

CHAPTER – 1

ACCIDENTS IN INDIA

Introduction

This is the tenth edition of the report after the comprehensive revision of the proforma for data collection of 'Accidental Deaths & Suicides in India' was done in 2014 in consultation with States/UTs Police and others stakeholders.

An accident is an unplanned event that results in injury, deaths, damage to property or some other losses. For statistical purpose, Accidental Deaths have been classified as deaths that could be due to forces/factors of the nature which have been termed as

'Accidental Deaths due to Forces of Nature' or could be due to deliberate or negligent conduct of human beings, which is termed as 'Other causes of Accidental Deaths'.

The rate of accidental deaths (per lakh of population) has increased in 2023 as compared to 2022. The total number of accidental deaths in 2023 is 4,44,104. [LIST-1.1]. The incidence and rate of accidental deaths during 2013-2023 is presented in Table-1.1.

LIST-1.1
Percentage Change in Population, Number and Rate of Accidental Deaths over the Corresponding Previous Year during 2019 to 2023

Year	Mid-Year Projected Population (In Lakh ⁺)	Percentage Change in Population over the Previous Year	Total Number of Accidental Deaths	Percentage Change in Accidental Deaths over the Previous Year	Rate of Accidental Deaths	Percentage Change in Rate of Accidental Deaths over the Previous Year
(1)	(2)	(3)	(4)	(5)	(6)	(7)
2019	13376.1 [#]	1.1	421104	2.3	31.5	1.3
2020	13533.9 ^{\$}	1.2	374397	-11.1	27.7	-12.1
2021	13671.8 ^{\$}	1.0	397530	6.2	29.1	5.1
2022	13797.5 ^{\$}	0.9	430504	8.3	31.2	7.2
2023	13923.3 ^{\$}	0.9	444104	3.2	31.9	2.2

[#] Source: Report of the Technical Group on Population Projections (November, 2019), National Commission on Population, Ministry of Health & Family Welfare.

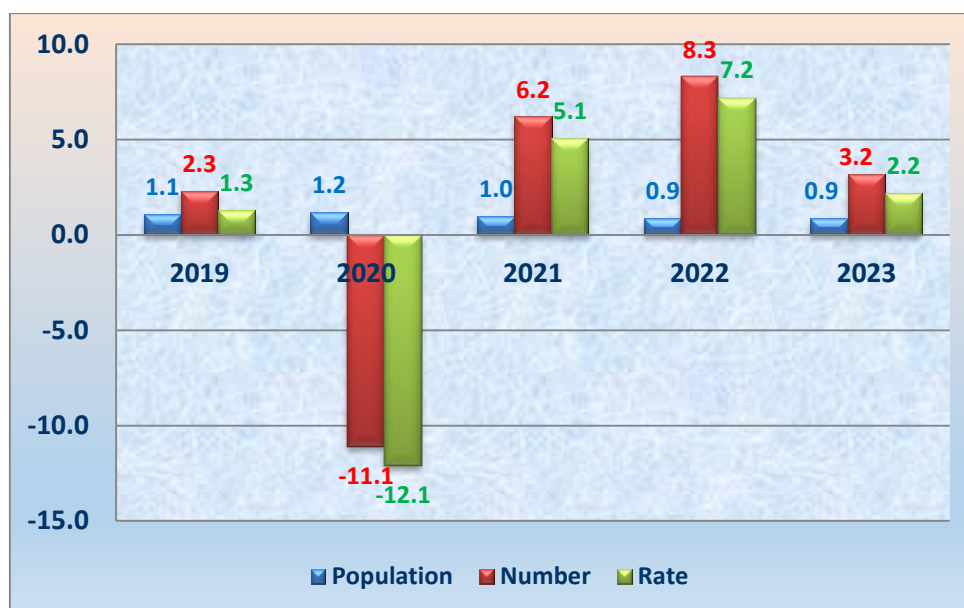
^{\$} Source: Report of the Technical Group on Population Projections (July, 2020), National Commission on Population, Ministry of Health & Family Welfare

⁺ One Lakh = 0.1 Million = 1,00,000

As per data provided by States/UTs

FIGURE-1.1

Percentage Change in Population, Number and Rate of Accidental Deaths during 2019 – 2023 (Over Corresponding Previous Year)



- As per data provided by

Out of 4,44,104 accidental deaths, 6,444 (1.5%) deaths were due to forces of nature and 4,37,660 (98.5%) deaths were due to other causes.

There was a decrease of 20.0% (from 8,060 deaths to 6,444 deaths) in deaths due to forces of nature and an increase of 3.6% (from 4,22,444 deaths to 4,37,660 deaths) in deaths due to other causes during 2023 as compared to 2022 **LIST-1.2 and Table 1.3.**

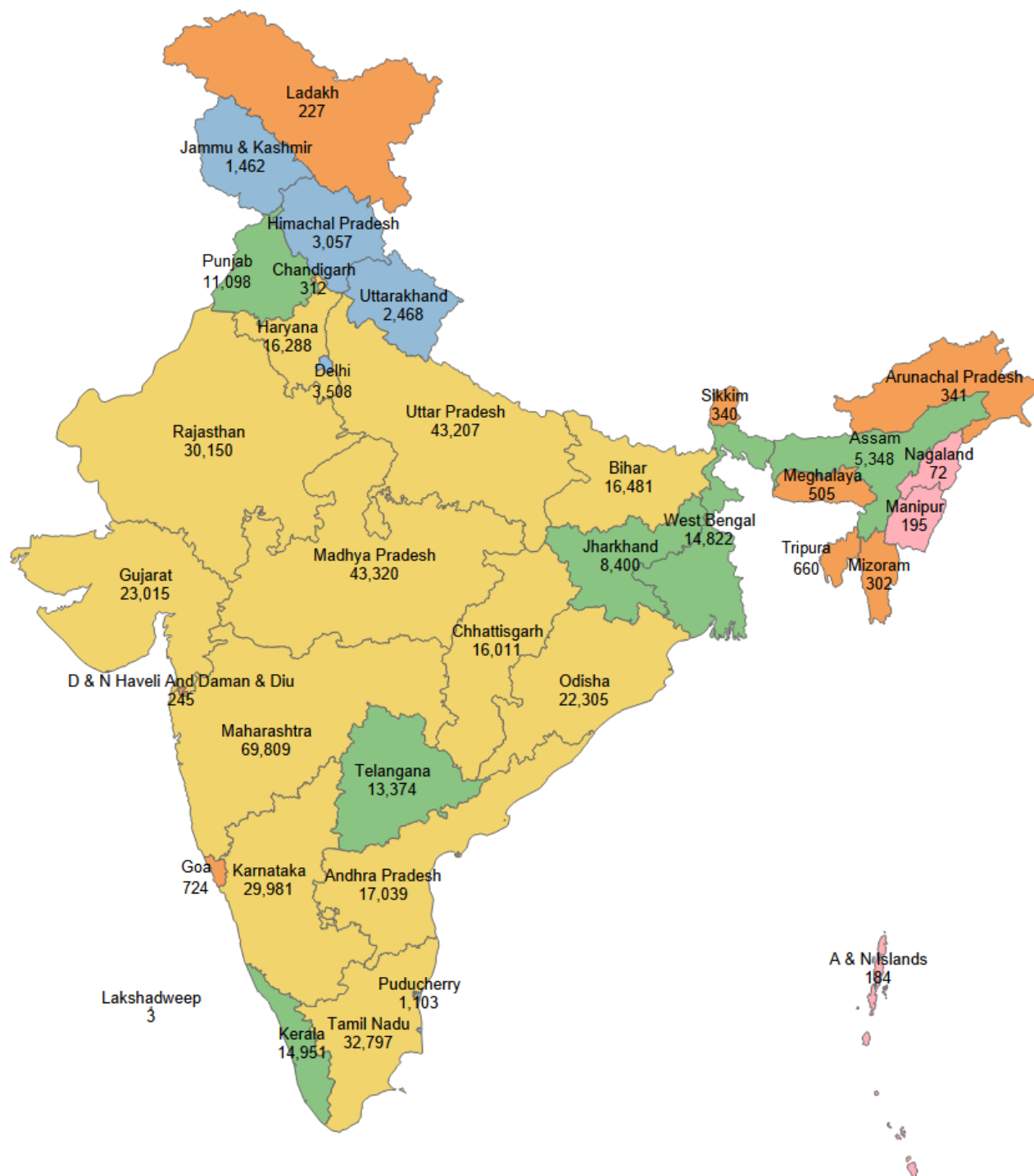
LIST-1.2

Number of Accidental Deaths due to Forces of Nature & Other Causes (2019 to 2023)

Year	Forces of Nature		Other Causes		Total Accidental Deaths	
	Number of Deaths	% change over previous year	Number of Deaths	% change over previous year	Number of Deaths	% change over previous year
(1)	(2)	(3)	(4)	(5)	(6)	(7)
2019	8,145	18.2	4,12,959	2.0	4,21,104	2.3
2020	7,405	-9.1	3,66,992	-11.1	3,74,397	-11.1
2021	7,126	-3.8	3,90,404	6.4	3,97,530	6.2
2022	8,060	13.1	4,22,444	8.2	4,30,504	8.3
2023	6,444	-20.0	4,37,660	3.6	4,44,104	3.2

- As per data provided by States/UTs.

FIGURE-1.2
STATE/UT – WISE ACCIDENTAL DEATHS DURING 2023



- As per data provided by States/UTs.

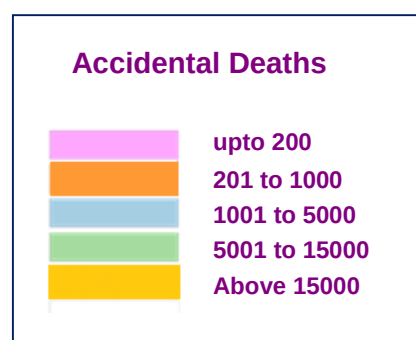
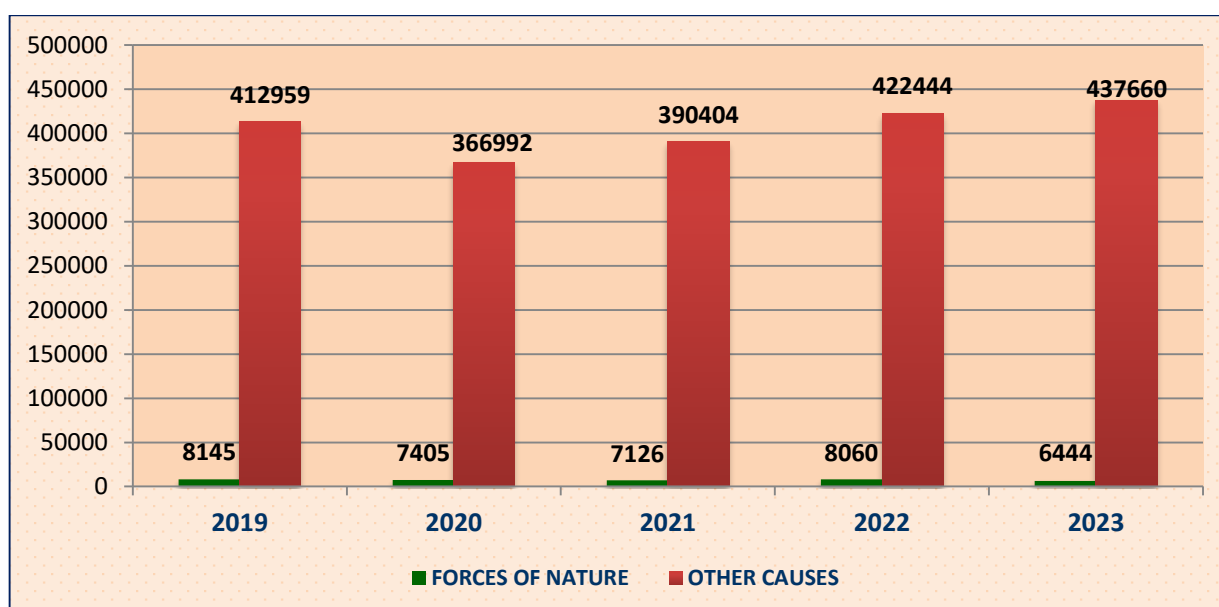


FIGURE-1.3

Total Number of Accidental Deaths due to Forces of Nature and Other Causes during 2019-2023



- As per data provided by

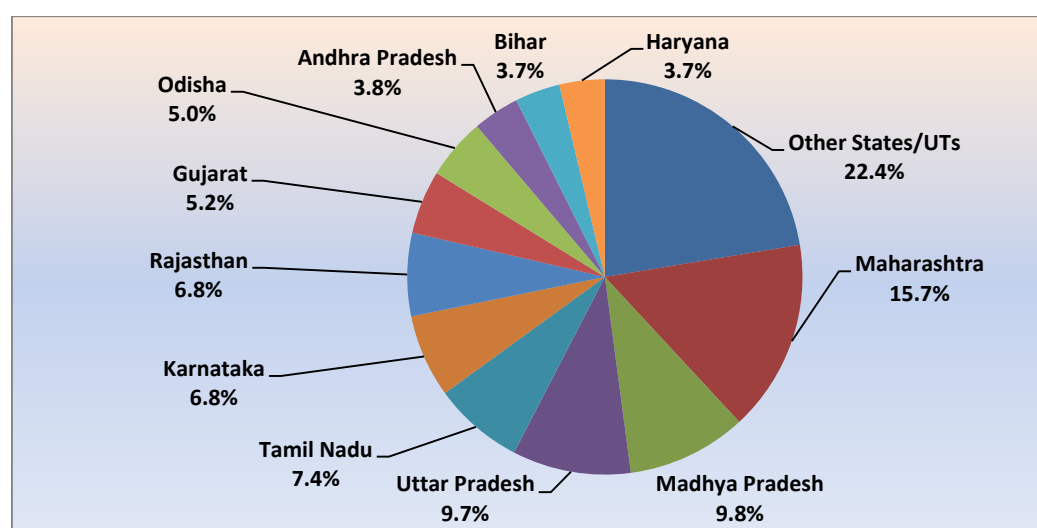
Number & Rate of Accidental Deaths in States/UTs

Maharashtra with a population share of 9.1% has reported highest number of accidental deaths (69,809), contributing (15.7%) of total accidental deaths reported in

the country. Uttar Pradesh, the most-populous State in the country with population share of 17.0% has accounted for (43,207) 9.7% of the total accidental deaths in the country. The other States having higher percentage share in total accidental deaths were Madhya Pradesh (9.8%), Tamil Nadu (7.4%), Karnataka & Rajasthan (6.8% each), Gujarat (5.2%), Odisha (5.0%), Andhra Pradesh (3.8%) and Bihar & Haryana (3.7% each) [Table-1.2].

FIGURE-1.4

Percentage Share of States/UTs in Accidental Deaths during 2023



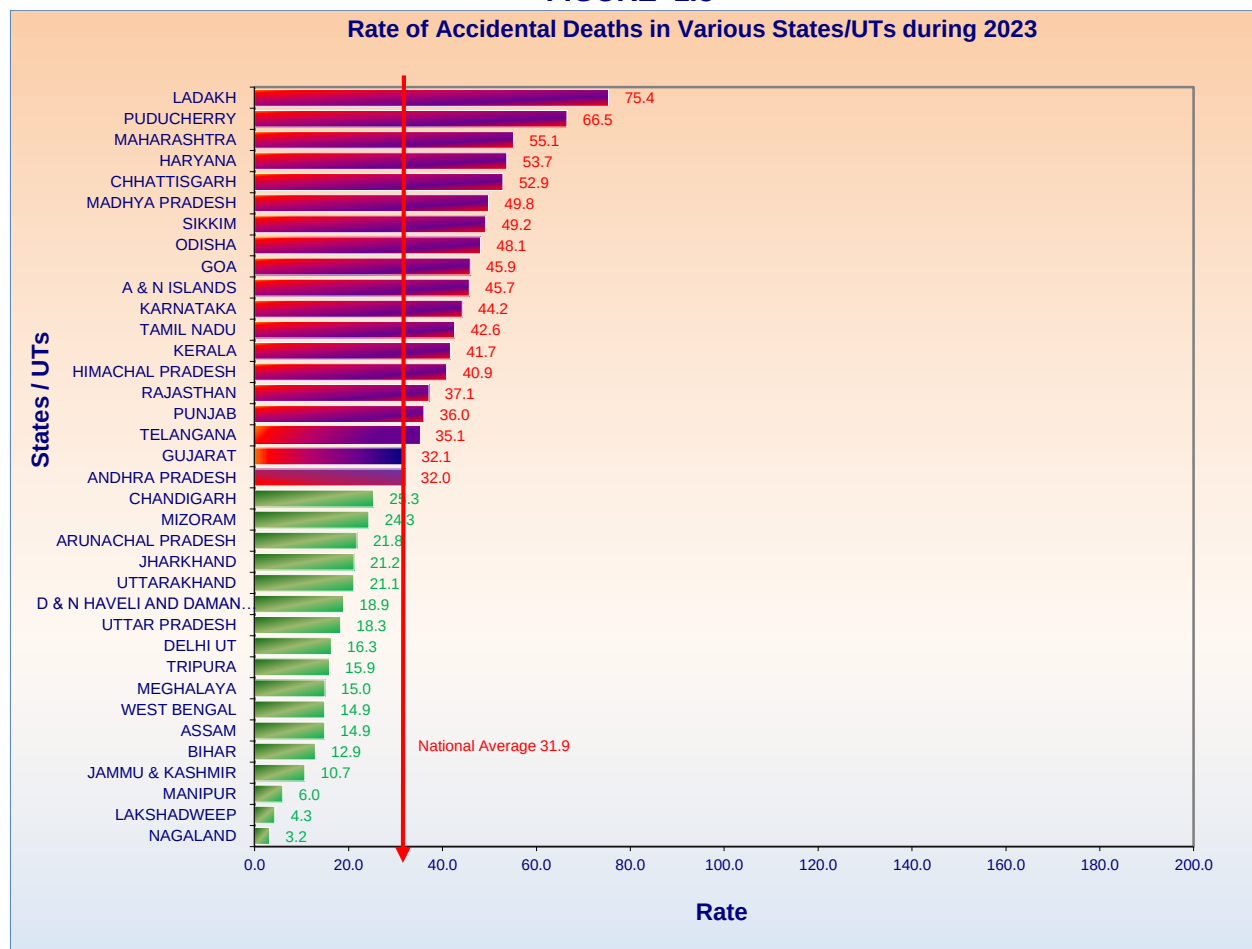
- As per data provided by

Other States/UTs include, Arunachal Pradesh, Assam, Chhattisgarh, Goa, Himachal Pradesh, Jharkhand, Kerala, Manipur, Mizoram, Meghalaya, Nagaland, Punjab, Sikkim, Telangana, Tripura, Uttarakhand, West Bengal, A & N Island, Chandigarh, D&N Haveli and Daman & Diu, Delhi, Jammu & Kashmir, Ladakh, Lakshadweep and Puducherry

The highest **rate of accidental deaths** was reported from Ladakh (75.4) followed by Puducherry (66.5), Maharashtra (55.1), Haryana (53.7), Chhattisgarh (52.9), Madhya Pradesh (49.8) Sikkim (49.2) and Odisha (48.1) against the national average of 31.9.

Nineteen—(19) out of 36 States/UTs have reported higher rates of ‘Accidental Deaths’ as compared to All India Average of 31.9 deaths per one lakh of population [**Figure – 1.5**].

FIGURE–1.5



- As per data provided by States/UTs.

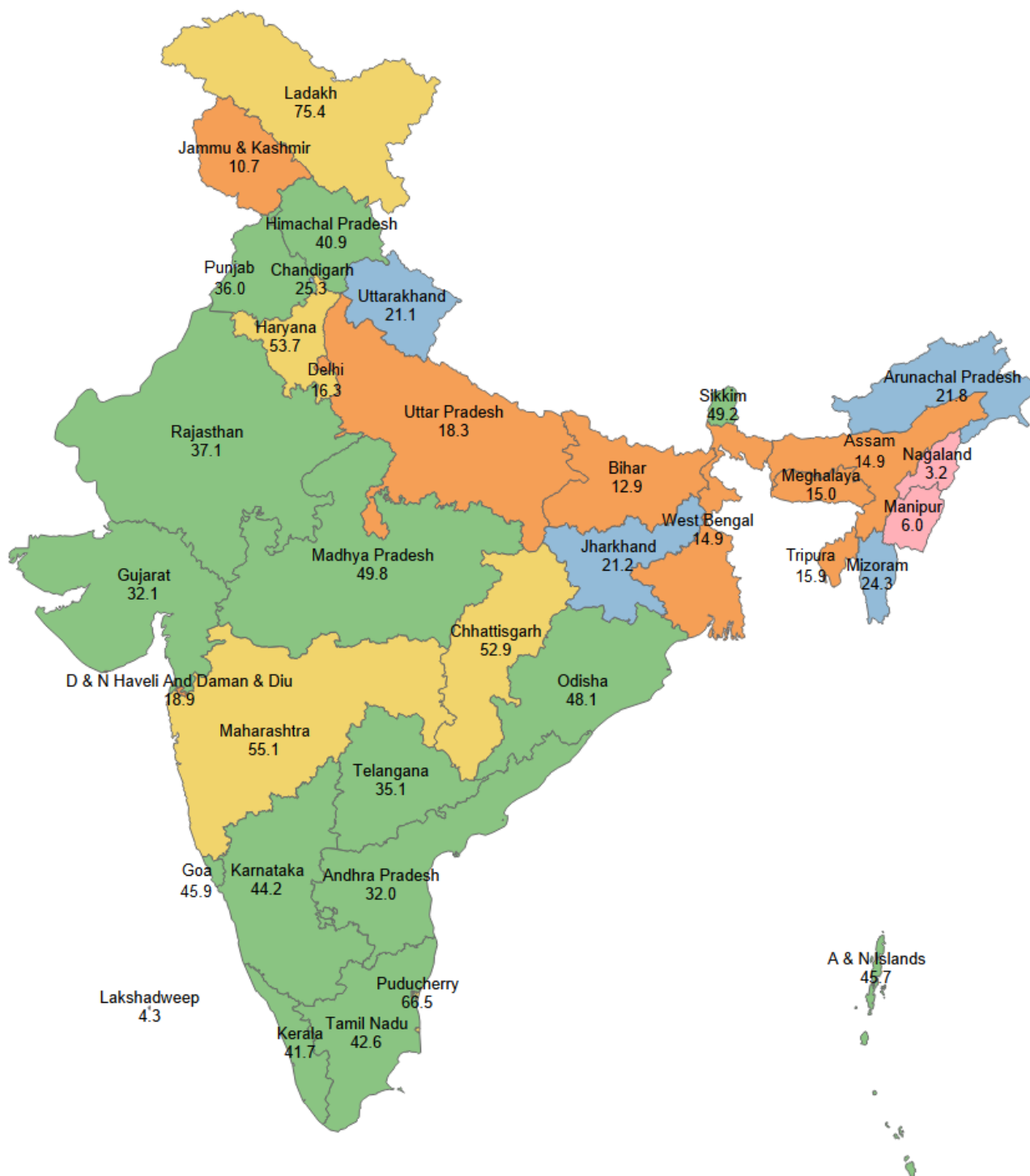
Among the States/UTs with significant number of accidental deaths; Rajasthan has reported the highest percentage increase (14.0%) in total accidental deaths in the year 2023 as compared to 2022 (from 26,449 in 2022 to 30,150 in 2023), followed by Tamil Nadu (12.6%), Jharkhand (8.1%), Himachal Pradesh (7.9%) and Odisha (6.1%).

Manipur has reported a decline of 44.8% in total accidental deaths in the year 2023 as compared to 2022 followed by D & N

Haveli and Daman & Diu (20.7%) and Arunachal Pradesh (16.6%).[**Table-1.3**]

Gender-wise analysis reveals that male and female constitute 82.1% and 17.9% of total victims respectively. Age group of majority of the victims was between 30 to 45 years. This group of people has accounted for 31.7% of all persons killed in accidents in the country during the year 2023. A total of 53,592 senior citizens (60 years & above of age) have died in various accidents during 2023 comprising 12.1%. [**Table-1.4**].

FIGURE-1.6
STATE/UT – WISE RATE OF ACCIDENTAL DEATHS DURING 2023



Note:

Rate of Accidental Deaths means Number of Accidental Deaths per 1,00,000 population.

- As per data provided by States/UTs.

Accidental Death Rate
All India Average 31.9

	Upto 10
	10.1 to 20
	20.1 to 30
	30.1 to 50
	Above 50

Causes of Accidental Deaths

Accidental deaths have been grouped into two broad categories i.e. 'Deaths due to Forces of Nature' and 'Deaths due to Other Causes'. Classification of types and causes of accidental deaths has been done for statistical purpose only.

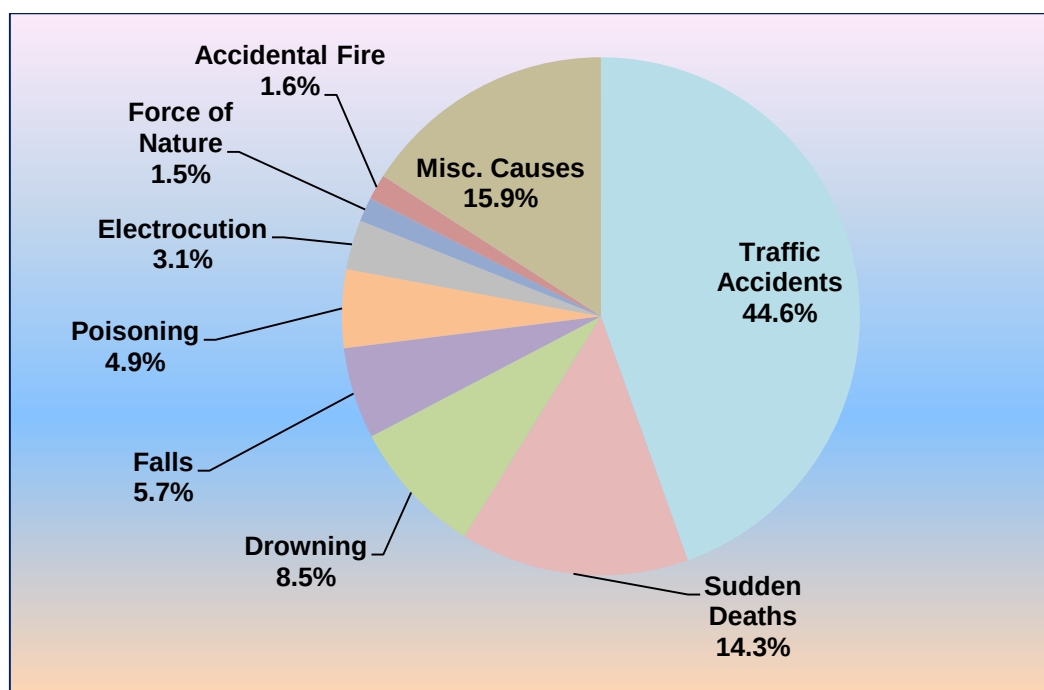
Under causes attributable to 'Forces of Nature', the deaths due to 'Heat/Sun Stroke', 'Exposure to Cold' and Tornado have increased during the year 2023 over the previous year 2022 whereas the deaths due to *inter-alia* 'Landslide', 'Lightning', 'Forest Fire', 'Avalanche', 'Torrential Rain', 'Flood' and 'Cyclone' have decreased in 2023 over 2022.

The deaths due to 'Stampede', 'Killed by animals', 'Accidental Explosion', 'Sudden Accidents', 'Electrocution', 'Deaths of Women During Pregnancy', 'Falls', 'Traffic Accidents' and 'Poisoning' have increased whereas the deaths due to Other Causes such as, 'Ship Accidents', 'Air Crash', 'Mines or Quarry disaster', 'Deaths Due to consumption of Illicit/Spurious Liquor', 'Suffocation', 'Firearm', 'Accidental Fire', 'Drug Overdose', 'Drowning' and 'Factory/Machine accidents' have decreased in 2023 as compared to the previous year 2022.

There has been an increase in number of traffic accidents deaths during 2023 (1,97,871 deaths) over during 2022 (1,94,347 deaths).

Cause-wise details of accidental deaths, its share and percentage variation during 2023 over 2022 are presented in **Table-1.0**.

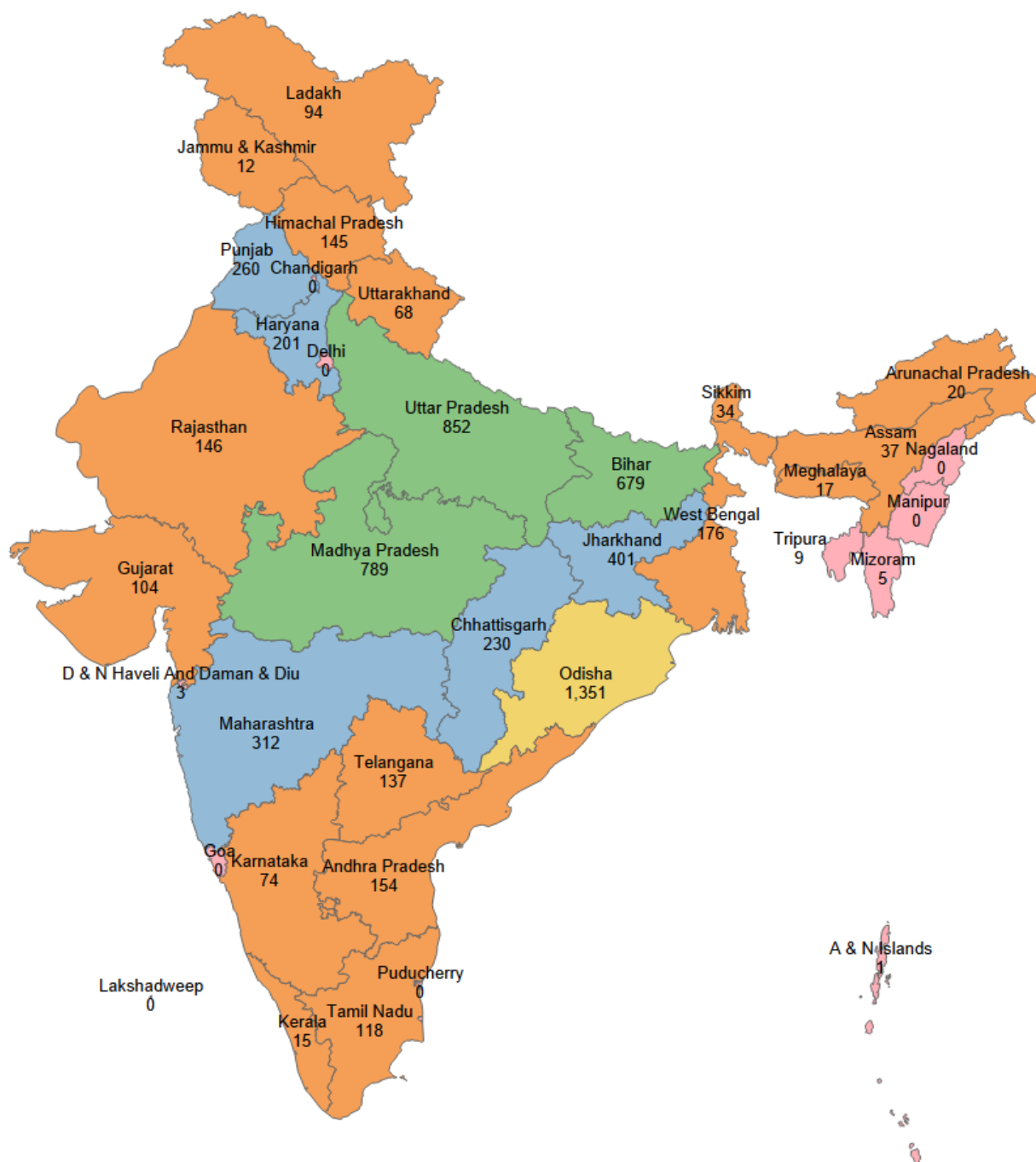
FIGURE-1.7
Percentage Share of Various Major Causes of Accidental Deaths during 2023
(Forces of Nature and Other Causes)



- As per data provided by States/UTs.

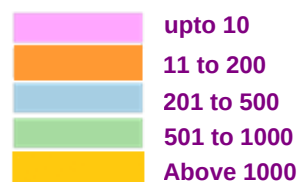
Misc. Causes include Collapse of Structure, Suffocation, Deaths of Women during Pregnancy, Killed by Animals, Deaths due to Consumption of Illicit/Spurious Liquor, Accidental Explosion, Drug Overdose, Factory/Machine Accidents, Firearm, Mines or Quarry Disaster, Stampede, Air-Crash, Ship Accidents, Other than above & Unknown Causes.

FIGURE-1.8
STATE/UT – WISE DEATHS DUE TO FORCES OF NATURE DURING 2023



- As per data provided by States/UTs.

Accidental Deaths



Accidental Deaths due to Forces of Nature

Cause - wise accidental deaths due to forces of nature are presented in **Table-1.0**. Out of 6,444 accidental deaths due to forces of nature, 39.7% deaths due to 'Lightning' (2,560 deaths), 12.5% deaths due to 'Heat/Sun Stroke' (804 deaths), 11.4% deaths due to 'Exposure to Cold' (733 deaths) and 4.1% deaths due to 'Flood' (266 deaths) during the year 2023. 'Landslide and 'Torrential Rain' caused 3.7% (239 deaths) and 0.9% (61 deaths) of total deaths due to forces of nature respectively.

The age group-wise distribution of accidental deaths due to forces of nature is given in **Table-1.5**. As per age-wise classification of deaths due to forces of nature, 3.2% of such victims were below 14 years of age (203 out of 6,444 deaths), 4.0% in the age-group of 14 years to below 18 years (257 deaths), 18.8% in the age-group of 18 years to below 30 years (1,213 deaths), 34.8% in the age-group of 30 years to below 45 years (2,242 deaths) and 28.8% victims were in age group 45 years to below 60 years (1,859 deaths) during 2023. Senior citizens (60 yrs. and above) constituted 10.4% (670 deaths) of the total such victims.

State/UT - wise and gender - wise distribution of accidental deaths in the country by forces of nature along with their share in total deaths is presented **Table-1.9**. 'Lightning' was the dominant factor and caused 100%, 95.9%, 81.3% and 78.3%, of total such deaths in D&N Haveli and Daman&Diu (3 out of 3 deaths), Karnataka (71 out of 74), West Bengal (143 out of 176 deaths) and Chhattisgarh (180 out of 230 deaths) respectively. 61.7% (124 out of 201 deaths) of death in Haryana and 48.8% (127 out of 260 deaths) of death in Punjab were due to 'Exposure to Cold'. 'Heat/Sun Stroke' was the dominant factor and caused 62.0% (85 out of 137 deaths) of total accidental deaths due to forces of nature in Telangana. 52.9% (18 out of 34 deaths) of deaths in Sikkim was due to 'Flood'. 'Landslide' was the dominant factor and caused 100.0%, 90.0%, 60.0%, 52.9% and 44.4% of total deaths in Mizoram (5 out of 5 deaths), Arunachal Pradesh (18 out of 20 deaths), Himachal Pradesh (87 out of 145 deaths), Meghalaya (9 out of 17 deaths)

and Tripura (4 out of 9 deaths) respectively during 2023.

Majority of deaths due to 'Torrential Rain' were reported in Meghalaya which accounted for 11.8% (2 out of total 17 deaths) of total such deaths in the country during 2023.

Accidental Deaths due to Other Causes

It is generally accepted that accidents due to 'Other Causes' are preventable and can be reduced by effective safety measures and safety consciousness.

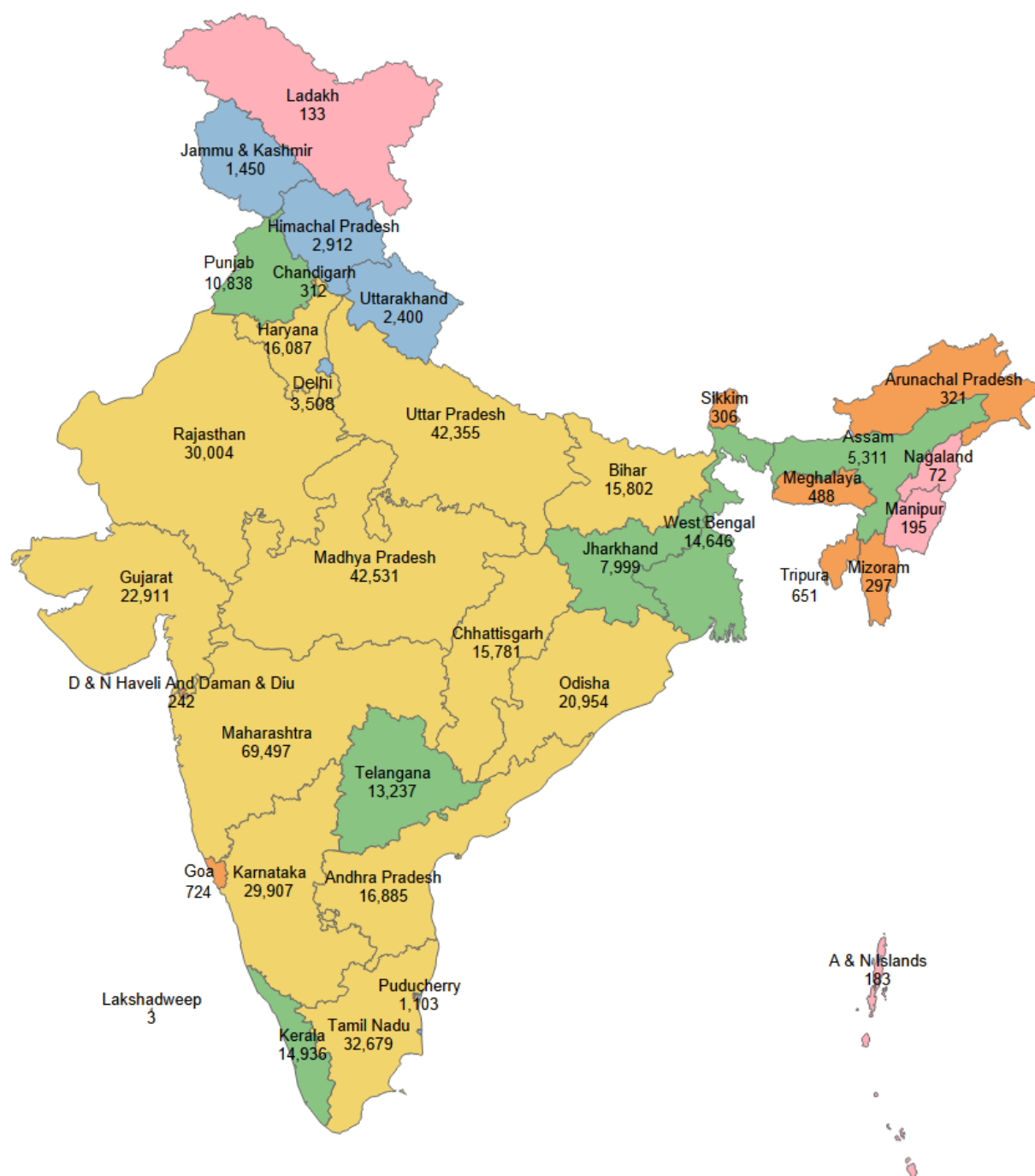
A total of 7,30,821 cases of Accidents due to 'Other Causes' were reported in the country which resulted in 4,37,660 deaths and 4,53,008 persons injured during 2023 [**Table-1.7**].

The cause-wise analysis of deaths revealed that majority of deaths were due to traffic accidents which accounted for 45.2% (1,97,871 out of 4,37,660 total deaths) of total deaths due to 'Other Causes' during the year 2023. Besides, 'Sudden Deaths', 'Drowning', 'Falls' and 'Poisoning' caused 14.5% (63,609 deaths), 8.6% (37,738 deaths), 5.7% (25,150 deaths) and 5.0% (21,785 deaths) of total deaths due to other causes respectively. [**Table-1.0**]

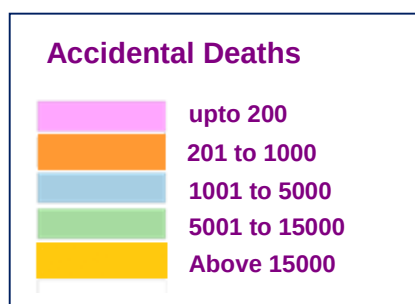
The age-group-wise distribution of accidental deaths in the country by various 'Other Causes' along with their share in total deaths is presented in **Table-1.8**. Majority of victims were in age groups 30 - below 45 years and 18 - below 30 years which accounted for 31.7% (1,38,691 deaths) and 24.5% (1,07,041 deaths) of total such deaths respectively.

State/UT-wise and gender-wise distribution of accidental deaths in the country by various other causes is presented in **Table-1.10**. Maharashtra with 9.1% of country's population has reported 15.9% (69,497 out of 4,37,660 deaths) of accidental deaths due to other causes. Madhya Pradesh, Uttar Pradesh, Tamil Nadu, Rajasthan, Karnataka and Gujarat accounted for 9.7%, 9.7%, 7.5%, 6.9%, 6.8% and 5.2% of total such deaths respectively during the year 2023.

FIGURE-1.9
STATE/UT – WISE ACCIDENTAL DEATHS DUE TO OTHER CAUSES
DURING 2023



- As per data provided by States/UTs.



State/UT-wise and Cause-wise details on 'Accidents due to Other Cause' is presented in **Table-1.10**. State/UT-wise analysis reveals that Uttar Pradesh accounted for maximum deaths (28,103 out of 1,97,871) under traffic accidents. Maharashtra accounted for maximum deaths (21,310 out of 63,609) under 'Sudden Deaths'. Madhya Pradesh accounted for maximum deaths (5,149 out of 37,738) under 'Drowning'. Maharashtra accounted for maximum deaths (4,563 out of 25,150) under 'Falls'.

A total of 456 incidents of consumption of illicit/spurious liquor caused 522 deaths in the country during 2023. States where various such deaths were reported include Jharkhand (194 deaths) followed by Karnataka (79 deaths), Bihar (57 deaths), Chhattisgarh (37 deaths), Punjab (33 deaths) and Himachal Pradesh (25 deaths).

Fire Accidents

Considering the number of fire accidents in the country and potential of the damage that can be caused by such accidents, the Bureau has also collected data on fire accidents according to places of occurrence for appropriate preventive strategies viz. 'Fire in School Buildings', 'Fire in Commercial Buildings', 'Fire in Residential / Dwelling

Buildings', 'Fire in Government Buildings', 'Fire in Mines', 'Fire in Factories', 'Accidental Fire in Trains' and 'Accidental Fire in Vehicles' [**Table-1.11**].

A total of 7,054 cases of fire accidents were reported in the country during 2023, showing a decrease of 6.8% as compared to 2022 (7,566 cases). 7,054 incidents of fire accidents caused injuries to 284 persons and 6,891 deaths during 2023. The place of Occurrence-wise analysis of fire accidents revealed that 54.2% of total deaths (3,738 out of 6,891) were reported in residential/dwelling buildings during 2023. [**Table 1.12**].

Large number of cases of fire accidents were reported in Odisha (1,032 out of 7,054), accounting for 14.6% of total such cases during 2023 [**Table-1.12**].

The State/UT-wise analysis of place of fire accidents revealed that 25 States & UTs have reported 50.0% or more deaths due to 'Fire in residential or dwelling building' during 2023. These States/UTs were Assam, Meghalaya, Mizoram, Tripura, A & N Island, Ladakh (100.0% each), Himachal Pradesh (90.9%), Karnataka (89.7%), Bihar (88.2%) and Arunachal Pradesh (83.3%) [**Table-1.12**].

Accidents in Cities

The term 'Mega City' refers to a city which has a population of 10 lakhs or more. As per the Population Census 2011, the status of mega city has been accorded to 53 cities.

A total of 65,897 accidental deaths were reported in 53 mega cities during 2023. Maximum numbers of accidental deaths (8,974 deaths) were reported in Mumbai accounting for 13.6% of total deaths, followed by Pune (5,054), Bengaluru (4,414), Delhi (3,508), Jaipur (2,641) and Surat (2,243).

Rate of accidental deaths in these 53 mega cities was higher (41.0) than all India rate of accidental deaths (31.9). The rate of accidental deaths was highest in Vasai Virar (115.2), Faridabad (114.0), followed by Pune (100.1), Raipur (95.6), Nagpur (88.8), Jaipur (85.9) and Jodhpur (82.0) as compared to average rate for the cities (39.2) [Table-1.2].

Out of 65,897 accidental deaths in 53 mega cities; 602 and 65,295 deaths were due to 'Forces of Nature' and 'Other Causes' respectively during 2023. A decrease of 10.1% was observed in deaths due to forces of nature and an increase of 4.8% was witnessed in deaths due to 'Other Causes' during 2023 over 2022 respectively. [Table-1.3]

Kolkata, the third largest city in terms of population (8.8% of the total population of all mega cities) among 53 cities, reported the low share of deaths in cities accounting for 0.4% of total accidental deaths in 53 mega cities during 2023. The comparative reporting of 'Accidental Deaths' during 2023 in the top four-metropolitan cities is presented in LIST-1.3.

Accidental Deaths due to Forces of Nature in Cities

A total of 602 deaths due to forces of nature were reported in these 53 mega cities during 2023. Among specified causes of accidents due to forces of nature, maximum deaths were reported due to 'Exposure to cold' followed by 'Heat/Sun stroke', accounting for 21.1% and 19.6% of total such accidental deaths. Due to 'Exposure to cold' maximum deaths were reported in Amritsar (51 deaths), Ludhiana (22 deaths) and Kanpur (15 deaths). Due to 'Heat/Sun stroke' deaths were reported in Amritsar (56 deaths), Ludhiana (20 deaths) and Dhanbad (11 deaths).

Accidental Deaths due to Other Causes in Cities

A total of 1,19,349 cases of 'Other' accidents were reported in 53 mega cities which caused injuries to 64,394 persons and 65,295 deaths during 2023. The cause-wise analysis of deaths due to other causes revealed that 29.0%, 27.4%, and 6.6% deaths in 53 cities were due to 'Sudden Deaths' (18,952), 'Traffic Accidents' (17,911), and 'Falls' (4,323) respectively.

Among 53 cities, Mumbai city (8,973) accounted for 13.7% of total deaths reported under 'Other causes' followed by Pune (5,053) and Bengaluru (4,414) which accounted for 7.7% and 6.8% respectively during 2023.

"Data on City-wise information is available on our website under 'Additional Tables for ADSI'"

LIST-1.3
Rate of Accidental Deaths in Metropolitan Cities during 2023

Sl. No.	City	Population (In Lakhs)	Share of Population In Cities (Percent)	Share of Deaths In Cities (Percent)	Rate (No. of Deaths Per One Lakh Population)
(1)	(2)	(3)	(4)	(5)	(6)
1.	Chennai	87.0	5.4	1.7	12.8
2.	Delhi	163.2	10.2	5.3	21.5
3.	Kolkata	141.1	8.8	0.4	1.7
4.	Mumbai	184.1	11.5	13.6	48.7

- As per data provided by States/UTs.
